Date: 12/07/2026

To
Controller of Patents
The Patent Office
I.P.O Building
Antop Hill,
S.M. Road
Mumbai - 400037

Kind Attn.: **Sukhdev Singh**, Controller of Patents

Re: Response to First Examination Report (FER)
Indian Patent Application No.: 202521045231.
Filing Date: 10/05/2025
Applicants: Shah Shashwat Nimesh and Others
Date of First Examination Report: 30/01/26
Last date for filing response to the examination report: 30/07/26

Respected Sir/Madam,

We write in respect of the First Examination Report issued on **30/01/26** with regard to Indian Patent Application No. **202521045231**.

Please find below our response to the objections raised by the Learned Controller:

Objections

Regarding Objection B (1): Inventive Step

The Controller had cited following documents in respect of his objections regarding inventive step of the present invention in the FER:

D1: IN202421078054A

D2: EP3377799B1

The Learned Controller contends that Claim(s) (1-4) lack(s) inventive step, being obvious in view of the disclosure in the documents cited above, and also lack(s) clarity, conciseness and definitiveness.

OUR SUBMISSIONS:

The Applicants respectfully submit that the claims have been amended in response to the observations raised in the First Examination Report. For addressing clarity, conciseness, and definitiveness, the inventive structural and functional features of the invention have been incorporated into the principal claim. The detailed description and drawings have been revised to provide clear support for the amended claims, and the specification has been updated to more clearly describe the structure, operation, and advantages of the invention.

The amended claims clearly define the three-armed symmetric base (15), plurality of neodymium magnets (3), screws (11), obstacle traversal mechanism, and vertical pipeline traversal capability, thereby distinctly defining the scope of protection sought.

The amended Claims 1 to 4 are novel, involve an inventive step, and are fully supported by the complete specification for the reasons stated below.

COMPARATIVE ANALYSIS:

After a thorough analysis of the cited documents with the present invention, it can be observed that the present invention is structurally and functionally distinct from D1 and D2, whether considered individually or in combination. Now, the applicants respectfully provide a detailed comparative analysis of the present invention with D1 and D2 in a tabular form below:

S.No.	Present Invention	D1:IN202421078054A
1.	The present invention discloses a pipeline traversing robot (100) comprising a base (15) having a **three-armed symmetric structure**, wherein each arm extends radially outward from a central hub at approximately 120 degrees relative to one another, configured to wrap around the exterior surface of a pipeline such that wheels (10) mounted at the terminal ends of each arm maintain simultaneous radial contact with the pipe	D1 discloses an IoT enabled robot for detecting gas pipe leakage (100) having a mobile robot body on which the processing unit (1), sensors and wheels (10) are mounted. D1 does not disclose any three-armed symmetric base or radial wrapping structure; the body of D1 simply sits on top of the pipe surface, relying on gravity for contact and operating exclusively on horizontal pipe surfaces.

	surface from multiple directions.	
2.	The present invention comprises a plurality of **neodymium magnets** (3) fixed on the underside of the base (15), preferably proximal to the terminal ends of each arm, generating magnetic adhesion force with the metallic pipe surface and maintaining inward radial contact pressure of the wheels (10), thereby enabling stable horizontal and vertical traversal.	D1 does not disclose any magnets or magnetic adhesion means. D1 has **no mechanism to grip or wrap around the pipe** and relies entirely on gravity for contact, making vertical traversal physically impossible irrespective of any modification to D1.
3.	The present invention comprises a **plurality of screws** (11) positioned at the terminal ends of each arm of the base (15), enabling manual **pre-deployment angular adjustment** of the arms relative to the central hub to accommodate pipes having diameters in the range of 150mm to 200mm.	D1 does not disclose any screws or any mechanism for angular adjustment of the robot body to accommodate pipes of varying diameters. D1's body is of **fixed configuration**.
4.	The present invention comprises a castor wheel (9) mounted on one of the three arms of the base (15), which, in combination with the three-armed symmetric structure, enables sequential obstacle crossing such that at least two arms maintain simultaneous grip on the pipe surface via the magnets (3) and wheels (10) while the third arm, aided by the castor wheel (9), **traverses pipe joints**	D1 discloses a castor wheel (9) that enables 360-degree rotational maneuverability of the robot body. However, D1 has **no mechanism to negotiate pipe joints, bends, or surface irregularities, causing the robot to stop and become stuck upon encountering such obstacles**, as D1 lacks any wrapping arms or sequential gripping arrangement.

	and bends with directional reorientation.	
5.	The present invention enables **stable traversal of vertically oriented metallic gas pipelines** by utilizing neodymium magnets (3) to generate magnetic adhesion and inward radial contact pressure between the wheels (10) and the pipe surface.	D1 was designed to **operate exclusively on horizontal pipe surfaces**. Upon encountering vertically oriented pipe sections, which are commonplace in industrial gas pipeline networks, the robot became completely inoperable.
6.	The present invention enables **traversal across pipe joints, bends, and other surface irregularities** through sequential engagement of the three arms of the base (15), ensuring continuous adhesion and traction throughout obstacle crossing.	D1 had no mechanism to negotiate pipe joints, bends, or surface irregularities, causing the **robot to stop and become stuck upon encountering such obstacles.**

S.No.	**Present Invention**	**D2: EP3377799B1**
1.	The present invention (100) is an external pipeline traversal robot wherein the **plurality of neodymium magnets** (3) are fixed on the underside of the base (15) and **generate magnetic adhesion force against the open exterior curved surface of the metallic pipeline**, maintaining inward radial contact pressure of the wheels (10) without any lateral enclosure.	D2 discloses a robot that **travels inside a pipeline** using tracked drive means and roller means equipped with magnetic adhesion means **generating a magnetic force betwe en the robot and the interior pipeline wall**, which has a natural lateral constraint not present on an exterior pipe surface.
2.	In the present invention, the base (15) comprises a **three-armed symmetric**	D2 does not disclose any three-armed radial wrapping base or exterior gripping arrangement.

	structure with arms extending radially at approximately 120 degrees from a central hub (15), wherein the magnets (3) on each arm generate inward radial gripping force sufficient for vertical climbing of an exterior pipe surface.	D2's magnetic adhesion means is configured to function with tracked drive means and roller means against an enclosed interior pipe wall for non-destructive structural inspection.
3.	The present invention comprises a **gas sensor (2) and temperature and humidity sensor (6)** connected to the processing unit (1), which processes the sensor data using a machine learning model deployed on the data store (13) to **determine the likelihood of gas leakage and transmit alerts to the monitoring and control center (12).**	D2 does not disclose any gas sensor, temperature and humidity sensor, machine learning model, or gas leak detection functionality; D2 is **limited to in-pipe travel and non-destructive testing for structural inspection.**
4.	The present invention comprises **screws (11) at the terminal ends of each arm of the base (15)** enabling manual **pre-deployment angular adjustment** to accommodate pipes having diameters in the range of 150mm to 200mm.	D2 does not disclose any screws or angular adjustment mechanism for accommodating pipes of varying diameters, as D2's roller-and-track based system operates within the **fixed bore of an enclosed pipeline.**

It can be observed from above comparisons, the present invention considerably
differs from the inventions in document D1 and D2.

Document D1 discloses an IoT enabled robot for detecting gas pipe leakage comprising a processing unit, gas sensor, temperature and humidity sensor, ultrasonic sensor, imaging device, wheels, castor wheel, monitoring and control center, and data store. While D1 provides gas

leakage monitoring and remote inspection functionalities, it neither discloses nor suggests a three-armed symmetric base configured to wrap around the exterior surface of a pipeline, magnetic adhesion using neodymium magnets, adjustable arm geometry using screws, vertical pipeline traversal capability, or a mechanism for traversing pipe joints and bends.

Document D2 discloses a pipeline inspection robot employing magnetic adhesion for movement within the interior of a pipeline. D2 operates in a fundamentally different environment and is directed towards internal pipeline inspection. D2 neither discloses nor suggests a three-armed wrapping configuration, screw-based diameter adjustment mechanism, external pipeline traversal, sequential obstacle crossing, nor continuous radial gripping of the exterior surface of a pipeline.

The present invention provides a unique combination of features including a three-armed symmetric base (15), neodymium magnets (3) for magnetic adhesion, screws (11) for accommodating pipes of varying diameters, and a traversal mechanism enabling movement across vertical sections, pipe joints, and bends while maintaining continuous adhesion and traction.

The inventive step of the present invention lies in the integration of magnetic adhesion, adjustable radial gripping, vertical traversal, obstacle crossing, and remote inspection functionalities into a single pipeline traversing robot. These features are neither disclosed nor suggested by D1 and D2, either individually or in combination.

Accordingly, the solution provided by the present invention cannot be regarded as obvious or anticipated in view of the cited references D1 and D2, either individually or in combination, by a person skilled in the art.

Therefore, the amended Claims 1 to 4 involve an inventive step under Section 2(1)(ja) of the Patents Act, 1970 and are fully supported by the complete specification. The dependent claims derive patentability from the inventive independent claim and further define additional technical features of the invention.

In view of the discussion, the Applicants respectfully request the Learned Controller to withdraw the objection under Section 2(1)(ja) relating to inventive step and allow Claims 1 to 4 of the present application.

Regarding Objection B (2): SUFFICIENCY OF DISCLOSURE:

The applicant acknowledges the examiner's observation that an abstract was not filed along with the complete specification. An updated abstract has been prepared in compliance with Section 10(1) of the Patents Act, 1970 read with Rule 13 of the Patents Rules, 2003, and is filed along with this response. The abstract provides a concise summary of the invention within the prescribed word limit and references Figure 1 as required.

Regarding Objection B (3): CLARITY AND CONCISENESS:

1) The applicant acknowledges the examiner's observations regarding Claims 3 and 4 and submits that both claims have been rewritten to clearly specify the technical means and structural features enabling the claimed functions. A revised detailed description has been incorporated to provide clear support for the amended claims.

2) Regarding Claim 3: The rewritten Claim 3 now specifies the complete obstacle traversal mechanism in technical detail, including the role of the ultrasonic sensor in automatic halting, the imaging device in transmitting visual data to the operator, remote manual control via a mobile application, the sequential three-armed gripping mechanism ensuring continuous pipe adhesion during traversal, the ground clearance enabling passage over surface irregularities, and the castor wheel enabling directional reorientation on pipe bends.

3) Regarding Claim 4: The rewritten Claim 4 replaces the term 'claws' with the correct component reference to screws (11) and now clearly specifies that the adjustment is manual, performed pre-deployment, involves angular repositioning of each arm relative to the central hub, accommodates pipes in the diameter range of 150mm to 200mm, and requires no specialized tools or expertise.

Amended Claims 3 and 4 are filed along with this response.

Regarding Objection B (4): DEFINITIVENESS:

The applicant acknowledges the examiner's observations regarding ambiguous phrases in Claims 1 and 2 and submits that all claims have been rewritten to remove ambiguity and clearly define the scope of the invention. The specific phrases flagged by the examiner have been addressed as follows:

Phrase Flagged	How It Has Been Addressed
"having a fidget spinner design"	Replaced with a precise geometric description: three arms extending radially outward from a central hub

	at approximately 120 degrees from one another, forming a three-armed symmetric structure.
"accommodation of pipes with varying diameters"	Replaced with a specific range: pipes having diameters in the range of 150mm to 200mm.
"fixed on the rear side of the base, maintaining sufficient distance from touching the pipes"	Replaced with: fixed on the underside of each arm of the base at a predetermined clearance from the pipe surface.
"robot in a position configured for environmental visual data acquisition"	Replaced with: mounted on the base and configured to capture image data of the surrounding pipeline environment for remote visual inspection.
"generate sufficient magnetic force for stable adherence to metallic surfaces" (Claim 2)	Replaced with a precise functional description specifying magnetic adhesion force against the exterior surface of metallic gas pipes, maintaining inward radial contact pressure of the wheels during both horizontal and vertical traversal.

Amended Claims 1-4 are filed along with this response.

PART-III: FORMAL REQUIREMENTS

1) Date and Signature of Applicant

The applicant acknowledges the examiner's observation that the complete specification and accompanying documents were signed by only one applicant. All documents filed along with this response carry the signatures of all five applicants, namely Shah Shashwat Nimesh, Shah Kruti Bhavesh, Gupta Manasvi Ganesh, Jobanputra Khushi Dharmendra, and Karani Ruhina.

2) Statement and Undertaking (Form 3 Details)

Statement & Undertaking is submitted in Form 3, stating that the applicants have not made any application for the same / substantially the same invention outside India.

3) Format of Drawings

The applicant acknowledges the examiner's observations regarding the format of drawings. All eight drawing sheets have been updated in compliance with Rule 15 of the Patents Rules, 2003. Each sheet now

carries the names of all applicants at the top left corner, the application number 202521045231 below the applicant names, the sheet number and total number of sheets at the top right corner, and the signature of the applicant at the bottom right corner. Updated drawings are filed along with this response.

In the view of above discussion, the applicants respectfully request the Learned Controller to consider our submissions and waive the said objection.

In view of the above submissions, the Applicants request you to kindly accept this response and proceed to grant a patent. The Learned Controller is respectfully requested to give an opportunity to be officially heard in the matter before taking any adverse decision in respect of captioned application.

We thank you in advance for your cooperation in this regard.

Yours faithfully

Shah Shashwat Nimesh
Shah Kruti Bhavesh
Gupta Manasvi Ganesh
Jobanputra Khushi Dharmendra
Karani Ruhina

Applicant

Enclosures:

1. Claims
2. Marked-up copy of claims
3. Updated specifications
4. Updated Drawings
5. Form 3
6. Abstract